

Chapter - 4

Lists and Tuples

• Lists

Python lists are containers to store a set of values of any data types.

friends = ["apple", "akash", "rahman", 7, false]

str() int() bool()
 ↓ ↓ ↓
 can store value of any datatype

• List Indexing

A list can be indexed just like string

such as

```
L1 = [7, 9, "Harry"]
```

```
L1[0]      # 7
```

```
L1[1]      # 9
```

```
L1[90]     # error
```

Unlike strings which are immutable, lists are mutable.

• List Methods

Consider the following list:

```
L1 = [1, 8, 7, 2, 21, 19]
```

- > L1.sort(): updates the list to [1, 2, 7, 8, 19, 21]
- > L1.reverse(): updates the list to [19, 21, 2, 7, 8, 1]
- > L1.append(8): add 8 at the end of the list
- > L1.insert(3, 8): This will add 8 at the index 3.
- > L1.pop(2): will delete element at index 2 and return value
- > L1.remove(21): remove element 21 from the list.

Teacher's Signature _____

• Tuples in Python

A tuple is immutable data type in Python.

such as

`a = ( )`

Empty tuple

`a = (1, )`

tuple with only one element needs a comma

`a = (1, 2, 3)`

tuple with more than one element

• Tuples Method

consider the following tuple:

`a = (1, 2, 3)`

> `a.count(1)`: a `count()` will return no. of 1 occurs in a

> `a.index(1)`: will return the index of first occurrence

Chapter - 4

Practice - Set

1) Write a Program to store seven fruits in a list entered by the user.

2) Write a Program to accept marks of 6 students and display them in sorted manner.

3) Check that ^{tuple} `a` type cannot be changed in Python.

4) Write a Program to sum a list with 4 numbers.

5) Write a Program to count the numbers of zeros in the following tuple:

`a = (7, 0, 8, 0, 0, 9)`